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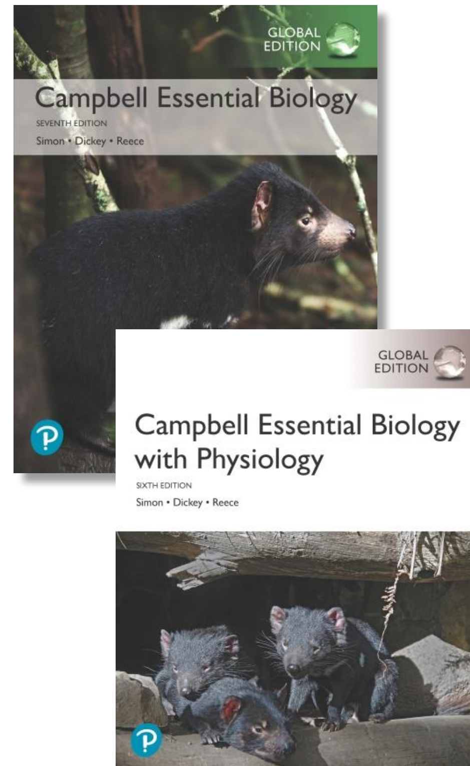

Chapter 29
The Working Plant

PowerPoint® Lectures created by Edward J. Zalisko, Eric J. Simon, Jean L. Dickey, and Jane B. Reece

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Like To Eat? Thank A Bacterium. Without Soil Bacteria, We’d All Starve.

Biology and Society: Planting Hope in the Wake of Disaster (1 of 2)

- The 2011 Fukushima and 1986 Chernobyl nuclear disasters left polluted soil and groundwater.
- Plants can help clean up polluted soil and groundwater in a process called phytoremediation.
  - Plants can efficiently move minerals and other compounds from the soil into the plant body.
  - Usually, the absorbed compounds are nutrients that help the plants grow.
  - But many species of plants, including sunflowers, are capable of absorbing large amounts of radioactive isotopes and heavy metals.

Phytoremediation: 식물이용 환경정화

Cleansing plants

Biology and Society: Planting Hope in the Wake of Disaster (2 of 2)

  - Cleanup crews have to remove only a few cubic meters of plant material rather than thousands of cubic meters of contaminated soil.
  - By using plants to clean up toxic wastes, we are benefiting from millions of years of plant evolution.
- Although plants can't move in search of food, they have an amazing ability to pull water and nutrients out of the soil and air.

How Plants Acquire and Transport Nutrients

- About 96% of a plant's dry weight is organic (carbon-containing) material synthesized from inorganic nutrients extracted from the plant's surroundings.
  - Plants obtain carbon dioxide (CO_2) from the air.
  - From the soil plants obtain water (H_2O) and **minerals**.
  - From the CO_2 and H_2O, plants produce sugars by photosynthesis. These sugars, combined with minerals, are used to construct all the other organic materials a plant needs.

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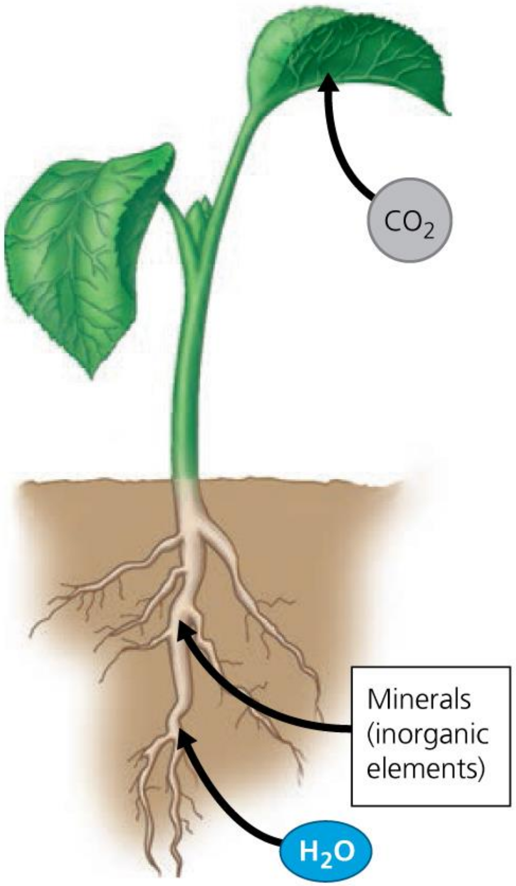
The uptake of nutrients by a plant



Plant Nutrition (1 of 4)

- A chemical element is considered an **essential element** if a plant must obtain it from its environment to complete its life cycle.
  - Seventeen elements are essential to all plants, and a few others are essential to some plants. Of the 17 essential elements,
    - 9 are **macronutrients**, required in relatively large amounts, and
    - 8 are **micronutrients**, which plants require in relatively small amounts.
  - **Figure 29.2** summarizes the nutrients that make up a typical plant body.

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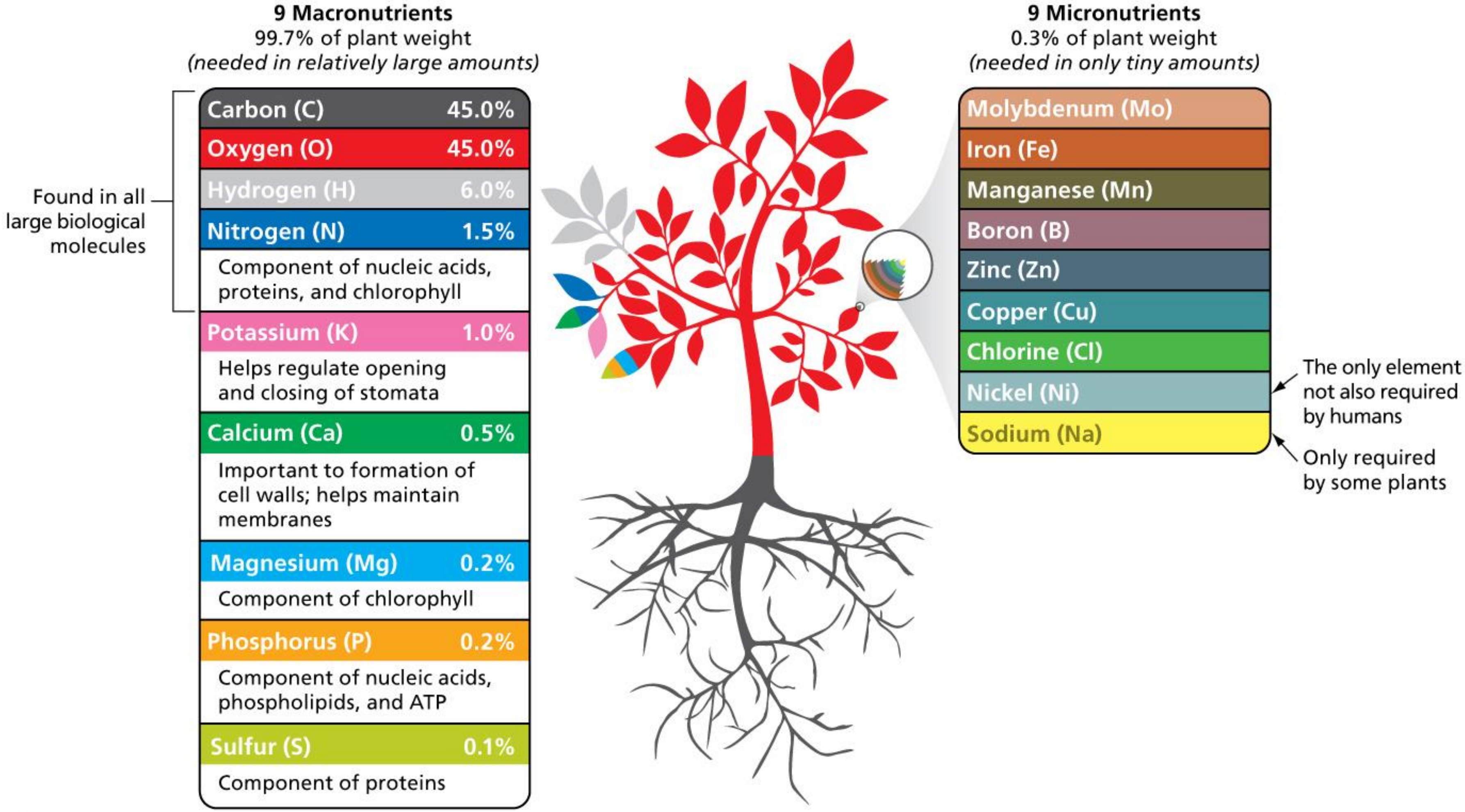
A summary of plant nutrients



Plant Nutrition (2 of 4)

- The quality of soil, especially the availability of nutrients, affects the health of plants and, for plants we consume, the quality of our own nutrition.
  - Nitrogen shortage is the most common nutritional problem for plants.
  - Nitrogen-deficient crop plants may produce grain, but the grain will have a lower nutritional value, and its nutrient deficiencies will be passed on to livestock or human consumers.

Plant Nutrition (3 of 4)

- **Fertilizers** are compounds that are applied to the soil to promote plant growth.
- There are two basic types of fertilizers.
  1. Inorganic fertilizers contain simple minerals, such as mined limestone (rich in calcium) or phosphate rock, or synthetic inorganic compounds such as ammonium nitrate.
     - Inorganic fertilizers come in a wide variety of formulations, but most emphasize the "N-P-K ratio," the relative amounts of the three nutrients most often deficient in depleted soils: nitrogen (N), phosphorus (P), and potassium (K).

Fertilizer

Plant Nutrition (4 of 4)

2. Organic fertilizers are composed of biologically derived products such as **compost**, a soil-like mixture of decomposed organic matter.

– Compost and other organic fertilizers are the key to **organic farming**. Organic farmers follow guidelines that promote sustainable agriculture which protects biological diversity, maintains and replenishes soil quality, manages pests with no (or few) synthetic pesticides, avoids genetically modified organisms, conserves water, and uses no (or few) synthetic fertilizers.

Compost: 퇴비, 두엄

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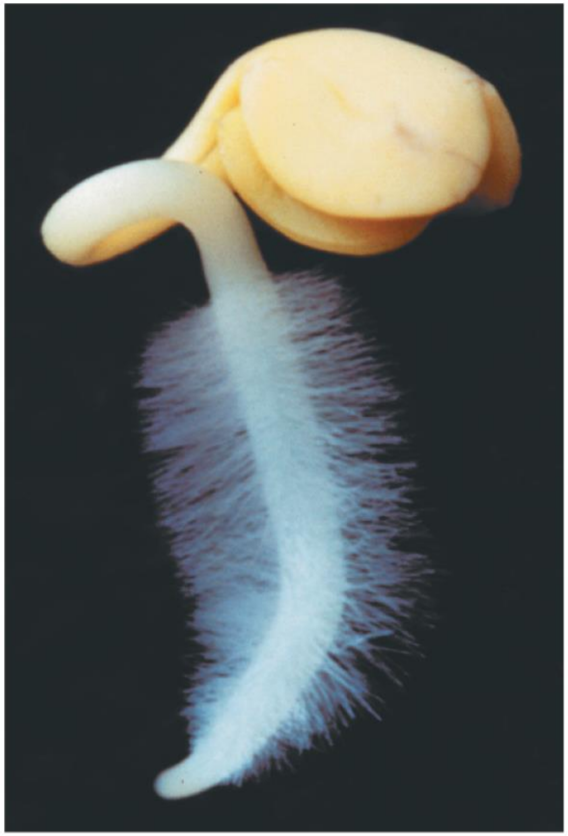
Compost

From the Soil into the Roots (1 of 3)

- Plant roots have a remarkable capacity for absorbing water and essential nutrients from soil because of their root hairs, extensions of epidermal cells that dramatically increase the surface area available for absorption.
  - All substances that enter a plant root are dissolved in water.
  - To be transported, water and solutes must move from the soil through the epidermis and cortex of the root and then into the water-conducting xylem tissue in the root's central cylinder.

Root hairs, which enable efficient absorption



From the Soil into the Roots (2 of 3)

- To reach the xylem, the solution must pass through the plasma membranes of root cells.
  - Because these membranes are selectively permeable, only certain solutes reach the xylem.
  - This selectivity helps regulate the mineral composition of a plant's vascular system.

From the Soil into the Roots (3 of 3)

- In an excellent example of interactions within biological systems, many plants gain significant surface area for absorption through symbiotic associations with fungi.
  - Together, the plant's roots and the associated fungus form a mutually beneficial (mutualistic) structure called a **mycorrhiza**.
  - Fungal filaments around the roots absorb water and minerals much more rapidly than the roots can alone.
  - Some of the water and minerals taken up by the fungus are transferred to the plant, while the plant's photosynthetic products nourish the fungus.

Mycorrhiza: 균근, 균뿌리

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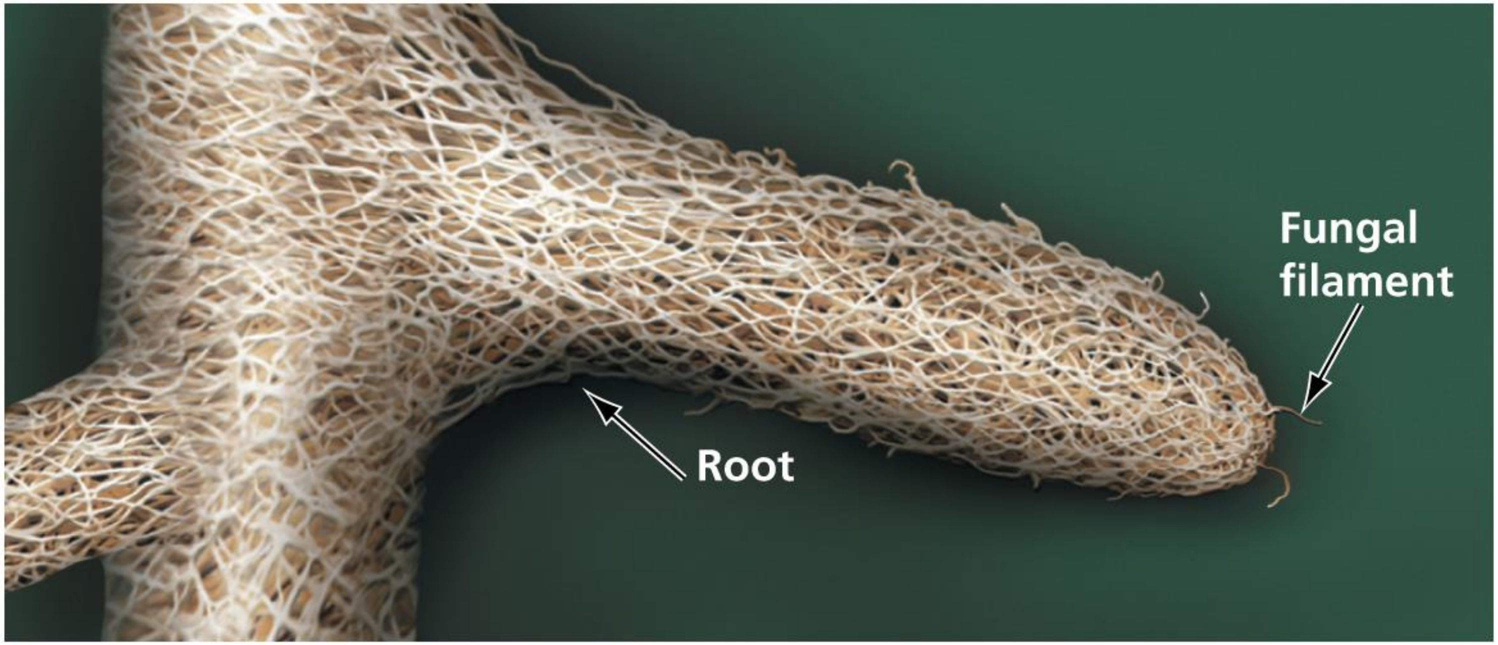
A mycorrhiza on a small tree root



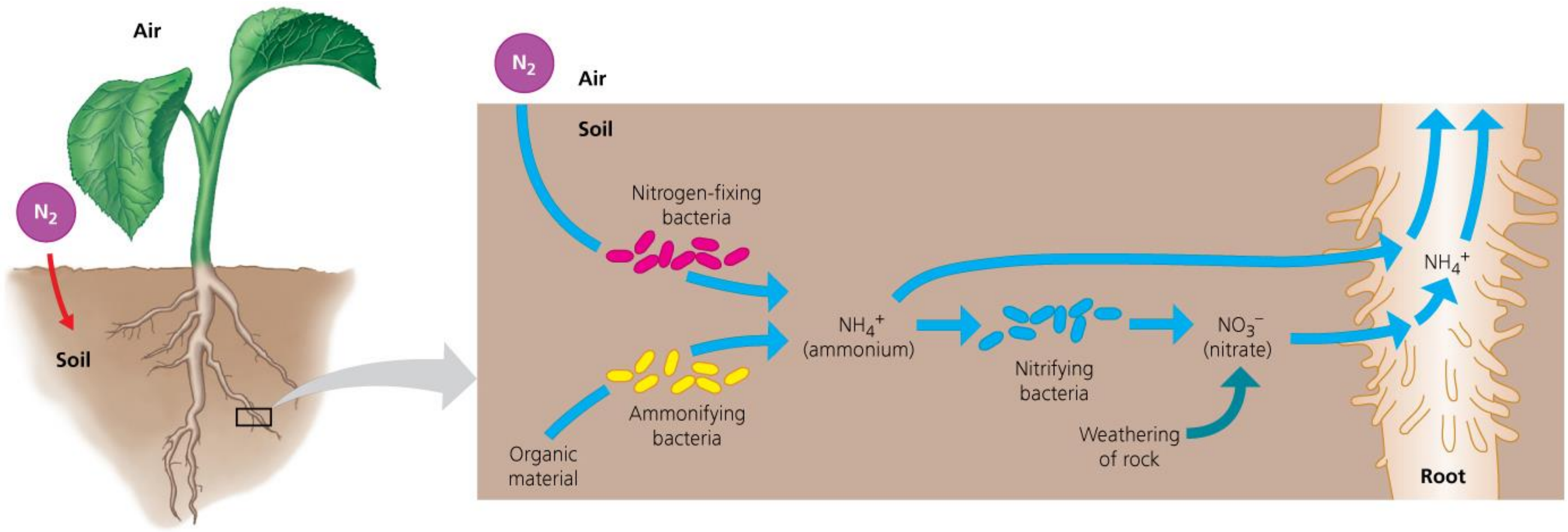
The Role of Bacteria in Nitrogen Nutrition

- To be useful, N_2 must first be converted to ammonium (NH_4^+) or nitrate (NO_3^-).
  - Such conversions illustrate the importance of transformations of energy and matter in living systems. In this instance, vital elements within the ecosystem are recycled, converted from one form to another.
  - A plant cannot perform this vital transformation, but bacteria can. We depend on bacteria: Without them, our crop plants could not obtain sufficient nitrogen, and we would be unable to feed ourselves.

Soil Bacteria and Nitrogen

- **Figure 29.7** shows three types of soil bacteria that play an essential role in supplying plants with nitrogen.
  1. Nitrogen-fixing bacteria convert atmospheric N_2 to ammonium, a process called **nitrogen fixation**.
  2. Ammonifying bacteria add to the soil's supply of ammonium by decomposing organic matter.
  3. Nitrifying bacteria convert soil ammonium to nitrate.

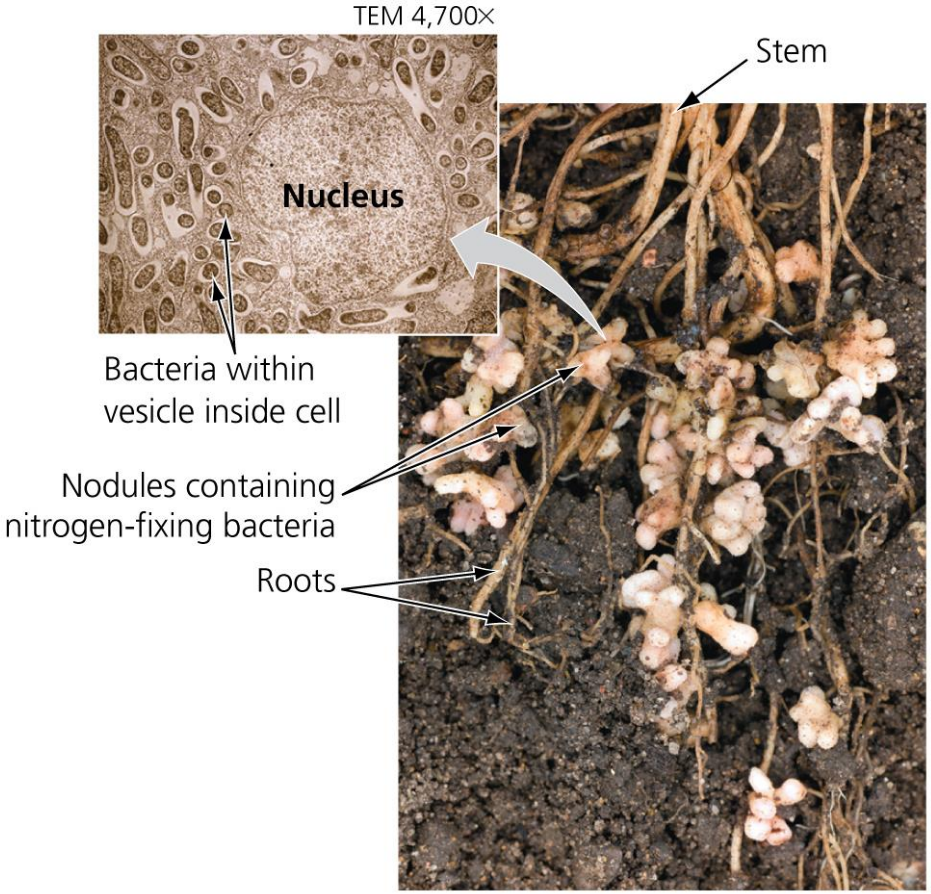
Figure 29.7 The role of bacteria in supplying nitrogen to plants



Root Nodule Bacteria and Nitrogen

- Some plant families, including legumes, such as peas, beans, peanuts, and other plants that produce their seeds in pods, have their own built-in source of ammonium: nitrogen-fixing bacteria that live in swellings called root nodules.
  - Within these nodules, plant cells have been "infected" by nitrogen-fixing bacteria that reside inside cytoplasmic vesicles.
  - The symbiotic relationship between a plant and its nitrogen-fixing bacteria is mutually beneficial. The bacteria have enzymes that help transform nitrogen; the plant provides the bacteria with other nutrients.

Root nodules of a pea plant containing nitrogen-fixing bacteria



The Transport of Water

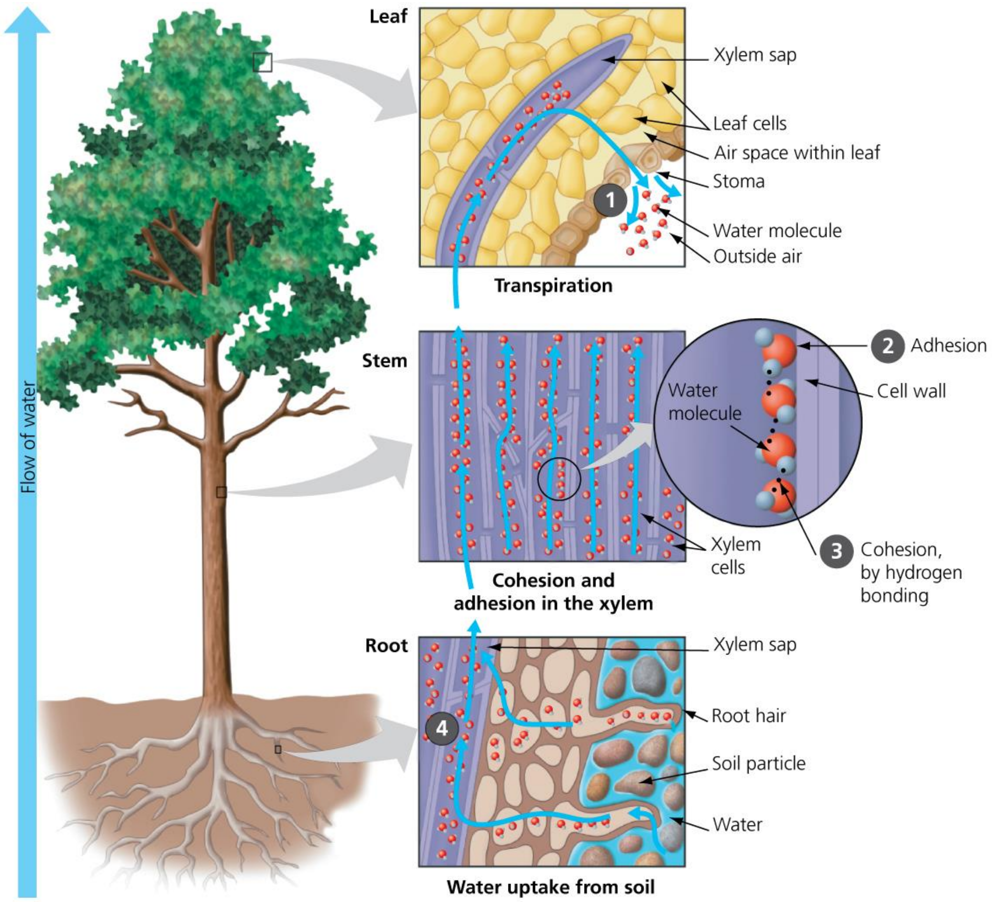
- To thrive, a typical tree must transport water from its roots to the rest of the plant.
  - In a mature plant, water-conducting cells of the xylem are arranged end to end to form very thin vertical tubes.
  - A solution of water and minerals, called **xylem sap**, flows through these tubes all the way from a plant's roots to the tips of its leaves.

The Ascent of Xylem Sap (1 of 2)

- What force moves xylem sap up against the downward pull of gravity?
  - Xylem sap is pulled upward by **transpiration**, the loss of water from the leaves of a plant by evaporation.
  - Most transpiration occurs through the stomata (pores) on the undersides of leaves.
  - **Figure 29.9** shows how transpiration pulls water up the xylem of a tree.

Transpiration: 증산

How transpiration pulls water up the xylem of a tree



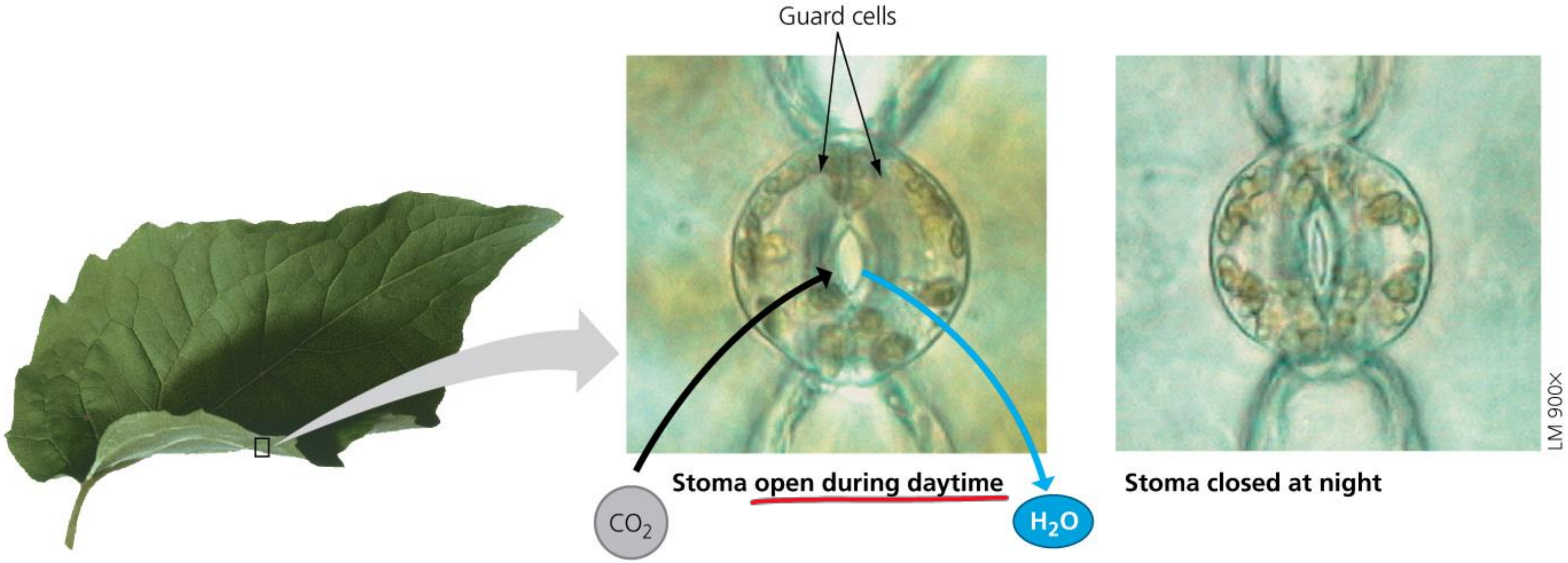
The Ascent of Xylem Sap (2 of 2)

- Transpiration can pull xylem sap up a tree because of two special properties of water.
  - **Adhesion** is the sticking together of molecules of different kinds.
  - **Cohesion** is the sticking together of molecules of the same kind.
  - Together, adhesion and cohesion create a continuous string of water molecules running from the roots to the leaves.
  - Biologists call this explanation for the ascent of xylem sap the **cohesion-tension hypothesis**.

The Regulation of Transpiration by Stomata

- Transpiration is needed to distribute water within a plant but can result in large amounts of water loss.
  - The leaf stomata, which can open and close, are evolutionary adaptations that help plants adjust their transpiration rates to changing environmental conditions.
  - The opening and closing of stomata are controlled by the changing shape of the two guard cells flanking each stoma.

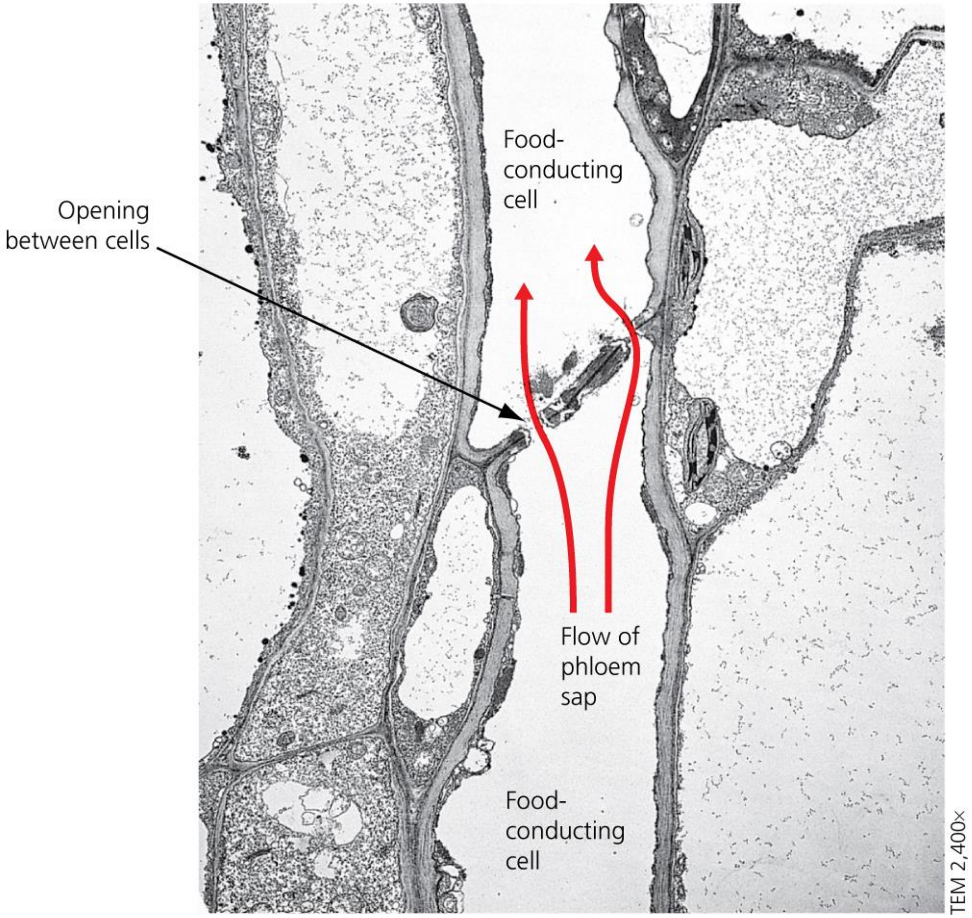
How guard cells around stomata control transpiration



The Transport of Sugars (1 of 2)

- A plant's survival depends on the transport of water and minerals absorbed from the soil and the transport of sugars it makes by photosynthesis. This is the main function of phloem.
  - Phloem consists of living food-conducting cells arranged end to end into long tubes. Through perforations in the end walls of these cells, a sugary liquid called **phloem sap** moves freely from one cell to the next.
  - **Phloem sap** contains inorganic ions, amino acids, hormones, and the disaccharide sugar sucrose.

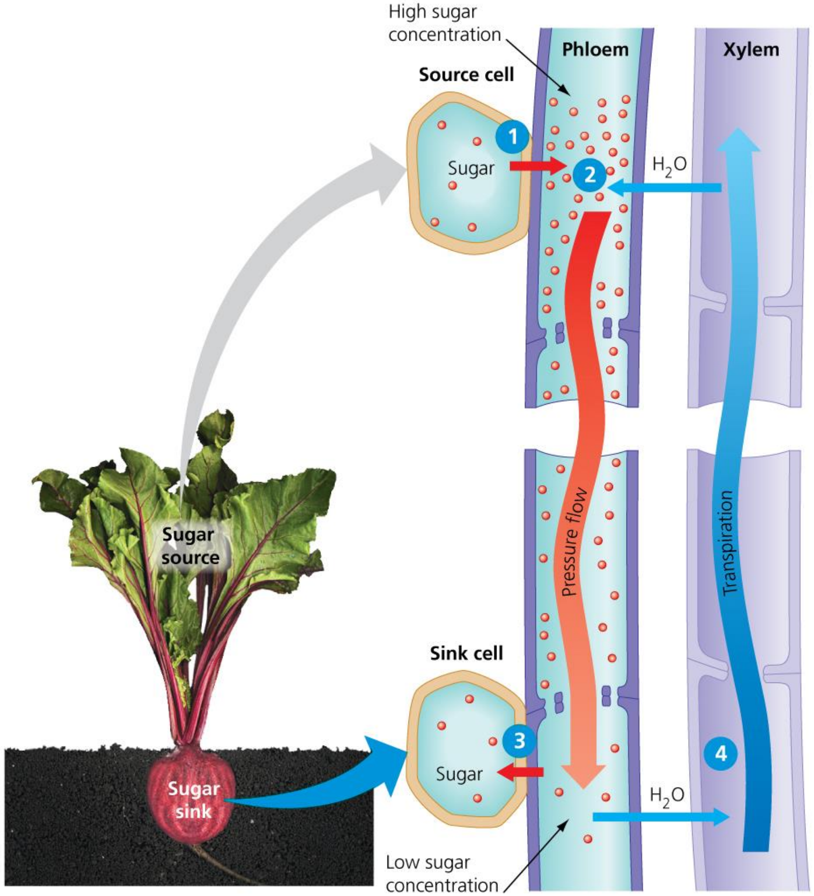
Food-conducting cells of phloem



The Transport of Sugars (2 of 2)

- Phloem sap moves in various directions from a **sugar source**, where sugar is produced, to a **sugar sink**, where sugar is stored or consumed.
  - Phloem sap moves from a sugar source to a sugar sink by the **pressure-flow mechanism**, illustrated in **Figure 29.12**.
  - The pressure flow mechanism explains why phloem sap always flows from a sugar source to a sugar sink, regardless of their locations in the plant.

Pressure flow in plant phloem



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The Process of Science: Can the Pressure Flow Mechanism Be Directly Measured?

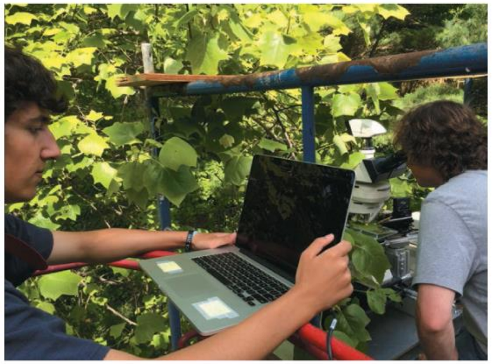
(1 of 3)

- **Background**: The pressure flow mechanism was originally proposed in 1930. But the technology that might provide definitive evidence did not exist. In recent years, a group of researchers have been working to develop the tools and techniques required to directly test this mechanism.
- **Method:** The only way to properly test the pressure flow mechanism is to measure fluid pressure within the phloem tubes of living plants. But how can this be accomplished without disrupting the system?

The Process of Science: Can the Pressure Flow Mechanism Be Directly Measured? (2 of 3)

- Michael Knoblauch, a plant cell biologist at Washington State University, may have found a way. His technique involves viewing a leaf from a living plant with a video microscope.
  - He then inserts a tiny glass needle— much thinner than a human hair—into phloem cells. The needle contains a minuscule amount of oil (about 0.0000000001 liter).
  - By measuring the extent to which the oil compresses as it is inserted into the cell, Dr. Knoblauch can calculate the pressure of the fluid inside the phloem.

Measuring the pressure flow mechanism



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The Process of Science: Can the Pressure Flow Mechanism Be Directly Measured? (3 of 3)

- **Results**: So far, these researchers have obtained several dozen measurements from within the leaves of morning glory plants and red oak trees.
  - By comparing pressure values—from near the sugar source versus far from it, for example, or within a short tree versus within a tall tree—they hope to obtain the evidence needed to either support or refute the pressure flow mechanism.
  - This is a great example of basic research: work that is largely concerned with building knowledge about the natural world.

Economic Uses of Plant Transport Products

- People have long taken advantage of plant transport mechanisms.
  - We harvest plant structures that are sugar sinks, such as storage roots (such as beets, carrots, and turnips) and tubers (such as potatoes and yams).
  - Maple syrup comes from starch that has been stored underground in sugar maple roots.
  - The sap of the desert dwelling *Aloe vera* is used as a soothing and moisturizing gel and in yogurt.
  - The rubber tree can be tapped to collect latex, also called natural rubber.

Plant Hormones

- A **hormone** is a chemical signal produced in one part of an organism and transported to other parts, where it acts on target cells to change their functioning.
  - Hormones provide plants with the ability to transmit information about the environment to various organs, thereby controlling responses that are essential to the life of the plant. Plants produce hormones in very small amounts.
  - Plant hormones control growth and development by affecting cell division, elongation, and differentiation.
  - **Table 29.1** lists five major types of plant hormones

Table 29.1 Major Types of Plant Hormones

Table 29.1	Major T ypes of Plant Hormones	
Hormone	Major Functions	Where Produced or Found in Plant
Auxins	Stimulate stem elongation; affect root growth, differentiation, and branching; stimulate development of fruit, apical dominance, phototropism, and gravitropism	Meristems of apical buds, young leaves, embryos within seeds
Cytokinins	Affect root growth and differentiation; stimulate cell division and growth; stimulate germination; delay aging	Made in roots and transported to other organs
Ethylene	Promotes fruit ripening; opposes some auxin effects; promotes or inhibits growth and development of roots, leaves, and flowers, depending on species	Ripening fruit, nodes of stems, aging leaves and flowers
Gibberellins	Promote seed germination, bud development, stem elongation, and leaf growth; stimulate flowering and fruit development; affect root growth and differentiation	Meristems of apical buds and roots, young leaves, embryos
Abscisic acid	Inhibits growth; closes stomata when water is scarce; helps maintain dormancy	Leaves, stems, roots, green fruit

Auxins (1 of 4)

- **Auxins** are a group of related hormones that are responsible for a wide range of effects on the growth and development of plants, such as promoting the elongation of cells.
  - **Phototropism** is the directional growth of a plant shoot in response to light. Such growth directs both seedlings and mature plants toward sunlight.
    - Microscopic observations of growing plants reveal the cellular mechanism that underlies phototropism.
    - Cells on the side of a stem that gets the least amount of light have elongated faster than cells on the other side, causing the shoot to bend toward the light.

Phototropism: growing toward light

Auxins (2 of 4)

- Auxins promote cell elongation in stems only within a certain concentration range. This same concentration range inhibits cell elongation in roots.
  - Above this range, auxins inhibit cell elongation in stems. Below this range, they cause root cells to elongate.
  - This complex set of effects reinforces two points:
    1. Different concentrations of a hormone may have different effects in the same target cell.
    2. A particular concentration of a hormone may have different effects on different target cells.

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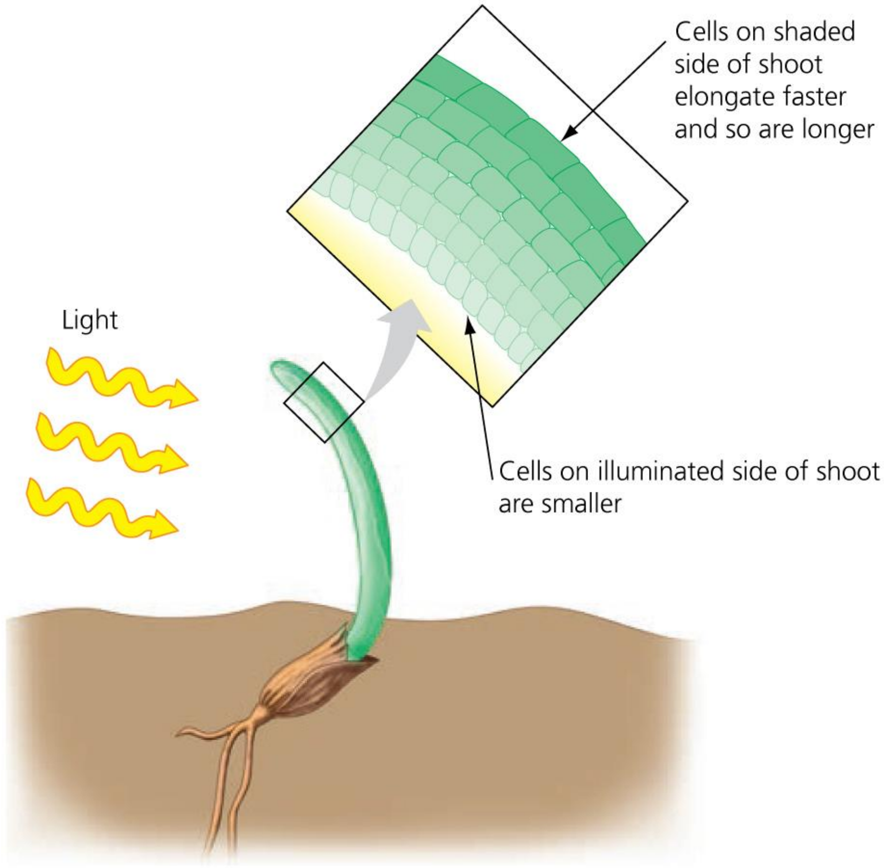
Cell elongation causes phototropism



Auxins (3 of 4)

- The use of synthetic plant hormones allows more food to be produced at lower cost.
  - One of the most widely used herbicides, or weed killers, is a synthetic auxin that disrupts the normal balance of hormones that regulate plant growth.
    - Because dicots are more sensitive than monocots to this herbicide, it can be used to selectively remove dicot weeds from a monocot lawn.
    - By applying herbicides to cropland, a farmer can reduce the amount of plowing required to control weeds, thus reducing soil erosion, fuel consumption, and labor costs.

Auxins (4 of 4)

- There is growing concern that the heavy use of synthetic chemicals in food production may pose environmental and health hazards.
  - Should we continue to produce cheap, plentiful food using synthetic chemicals and tolerate the potential problems?
  - Or should we put more of our agricultural effort into organic farming, recognizing that foods may be less plentiful and more expensive as a result?

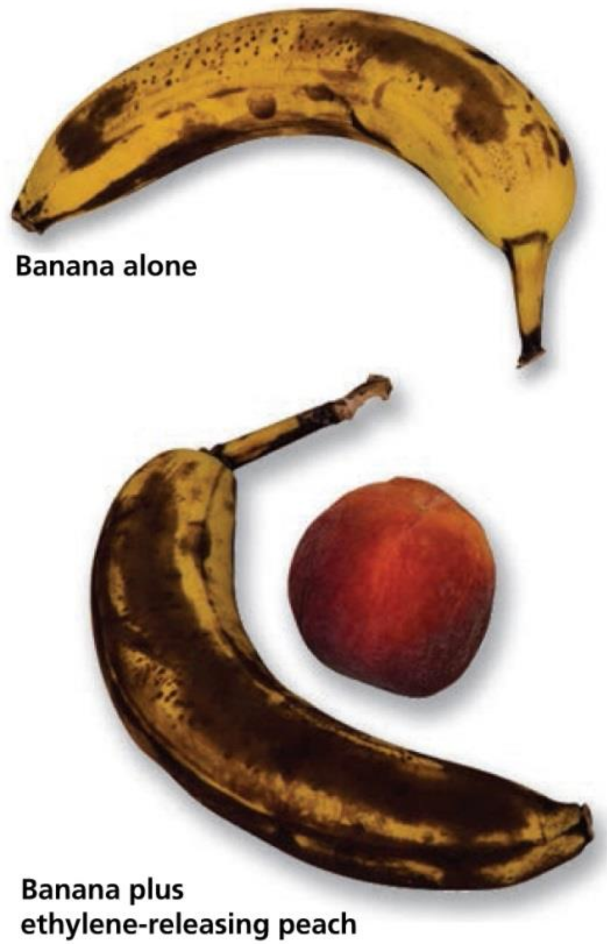
Cytokinins

- **Cytokinins** are a group of closely related hormones that act as growth regulators that promote cell division and are produced in actively growing tissues.
  - Cytokinins entering the shoot system from the roots counter the inhibitory effects of auxins coming down from the terminal buds.
  - The complex growth patterns of most plants probably result from the relative concentrations of auxins and cytokinins.
  - The interactions of biological components drive many of life’s processes.

Ethylene

- **Ethylene** is a hormone, released as a gas, that triggers a variety of aging responses in plants.
  - A burst of ethylene production in a fruit triggers its ripening.
  - Leaf drop in autumn is triggered by environmental stimuli, including the shortening days and cooler temperatures of autumn.
    - These stimuli cause a change in the balance of ethylene and auxins that weakens the cell walls in a layer of cells at the base of the leaf stalk.
    - The weight of the leaf, often assisted by wind, causes this layer to split, releasing the leaf.

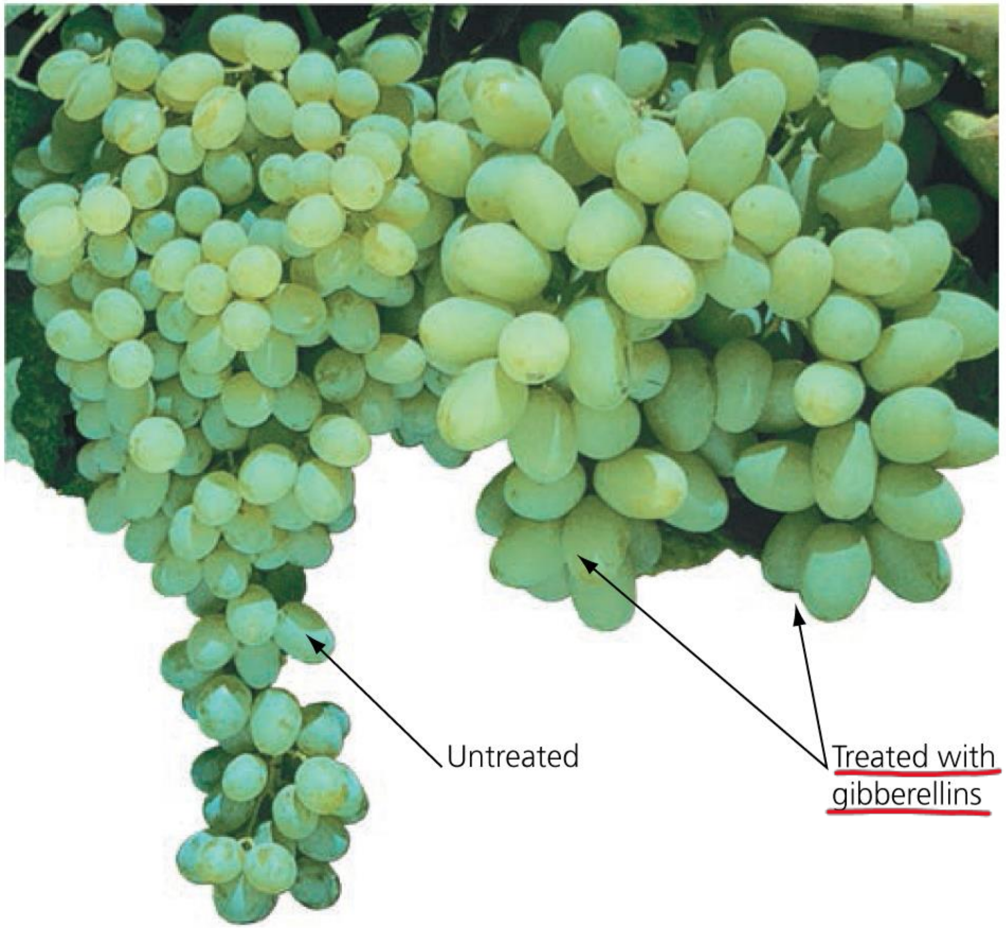
Effect of ethylene on the ripening of bananas



Gibberellins (1 of 2)

- **Gibberellins** are growth-regulating plant hormones which stimulate cell elongation and division in stems and leaves.
  - This action generally enhances that of auxins.
  - Also in combination with auxins, gibberellins can influence fruit development. For example, they are used commercially to make seedless grapes grow larger and farther apart in a cluster.

Effect of gibberellins on grapes



Gibberellins (2 of 2)

- Gibberellins are also important in seed germination in many plants.
  - Many seeds that require special environmental conditions to germinate, such as exposure to light or cold, will germinate when sprayed with gibberellins.
  - In nature, gibberellins in seeds are probably the link between environmental cues and the metabolic processes that reactivate growth of the embryo after a period of dormancy.

Abscisic Acid

- **Abscisic acid** *slows* growth and can suspend growth at the onset of seed dormancy.
  - Seed dormancy is an evolutionary adaptation that ensures that a seed will germinate only when the conditions of light, temperature, and moisture are appropriate for growth.
  - Many types of dormant seeds will germinate only when abscisic acid is removed or inactivated.
  - For many plants, the ratio of abscisic acid (which inhibits seed germination) to gibberellins (which promote seed germination) determines whether the seed will germinate or remain dormant.

Response to Stimuli: Tropisms

- Plants can respond to physical stimuli from the environment, including light, touch, and gravity. **Tropisms** are directed growth responses that cause parts of a plant to grow toward or away from a stimulus.
  - **Phototropism** is the directional growth of a plant shoot in response to light.
  - **Thigmotropism** is growth in response to touch, as when a pea tendril contacts a string or wire and coils around it for support.
  - **Gravitropism** is the directional growth of a plant organ in response to gravity.

Tropisms: directed growth responses

Phototropism	Thigmotropism	Gravitropism
 These seedlings bending toward the light are young cilantro plants (a common herb).	 Thigmotropism, growth in response to touch, allows vining plants to wrap around support structures. In this photo, the tendril of a pea plant coils around a wire fence.	 These seedlings were both germinated in the dark. The one on the left was left alone. The one on the right was turned on its side two days after germination so that the shoot and root were horizontal. When this photo was taken, the shoot had turned back upward and the root had turned down.

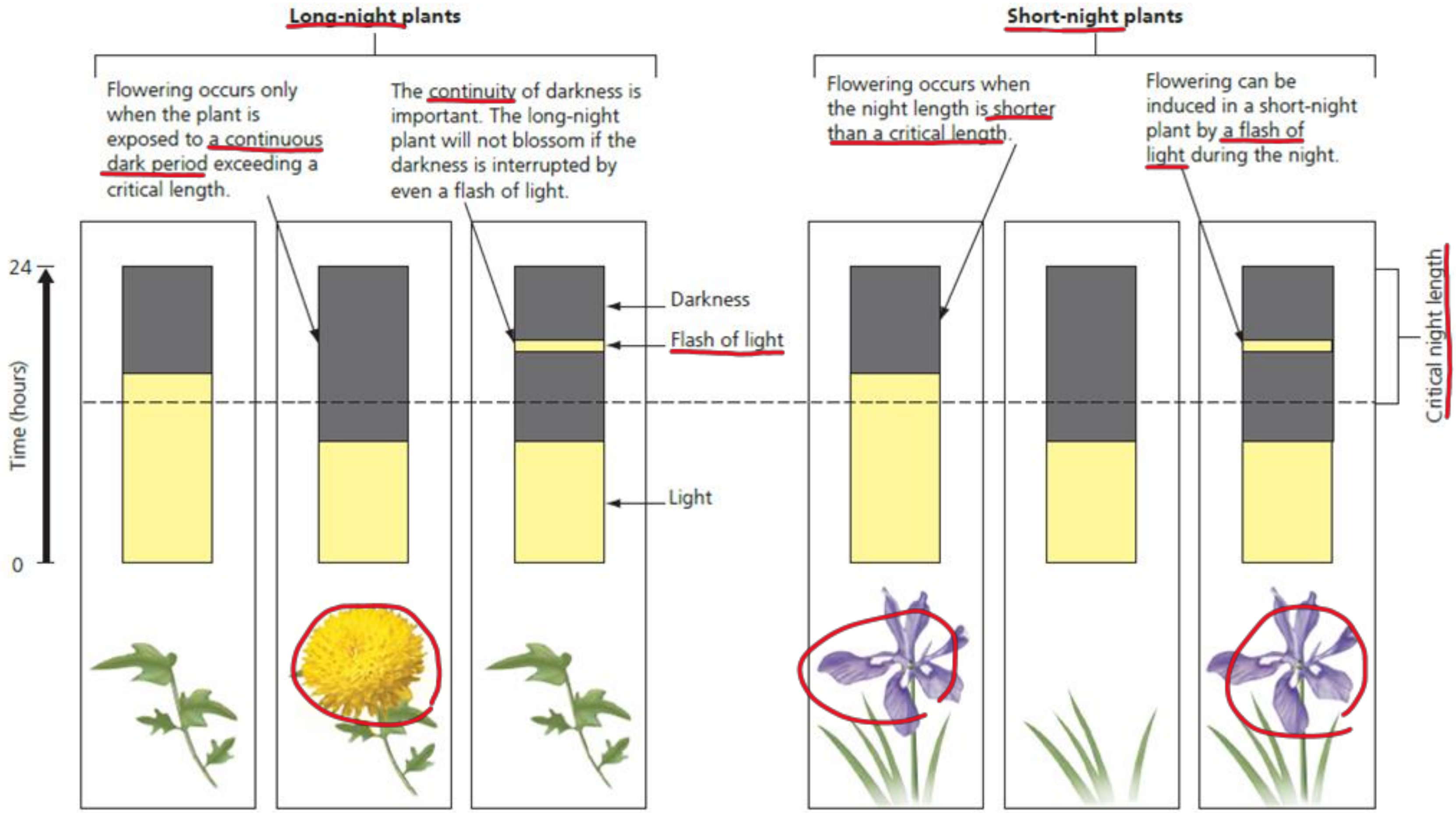
Photoperiodism (1 of 2)

- Light provides energy for photosynthesis, directs growth, and regulates a plant's life cycle.
  - Flowering, seed germination, and the onset and end of dormancy are stages in plant development that usually occur at specific times of the year.
  - The environmental stimulus that plants most often use to detect the time of year is the change in day length (photoperiod) or night length.
  - A response to specific day length or night length is called **photoperiodism**, which often affects flowering.

Photoperiodism (2 of 2)

- Originally, plant biologists referred to these plants as short-day and long-day plants. We now know that it is night length, not day length, that affects flowering.
  - Long-night plants generally flower in fall or winter, when nights lengthen.
  - Short-night plants usually flower in spring or summer, when nights are briefer.
- Some plants are night-neutral.

Photoperiodism and flowering



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